The International HPC Summer School

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What is the IHPCSS?

- The International High Performance Computing Summer School is a one-week, intensive, in-person training event held annually since 2010 at locations around the world.
- International partners like Access (US), EuroHPC (EU), RIKEN (Japan), Pawsey (Australia), and CHPC (South Africa) select students with strong potential in HPC.
- Partners fund student participation, instructors, and staff.
- The event includes a unique mentoring component to help students build careers in HPC.
- Compute Canada was a partner from 2014–2017, SciNet from 2018–2023, and the Alliance from 2025 onward as a Pilot.

NTCC IHPCSS Working Group

- Being a partner involves:
  - attending international organizing committee meetings
  - contributing to program development
  - running the Canadian selection process
  - inviting selected students
  - supporting them during the event
- The Alliance delegated these tasks to the National Training Coordination Council (NTCC), which started a working group (WG).
- The Alliance required a pilot funding proposal for the event, which the WG prepared. (It helped that the Alliance Training Coordinator is part of the WG).
- Finance and expense conditions were handled by the Alliance in 2025, and by SciNet in 2026 (requiring another proposal to the Alliance.)

IHPCSS 2025: Lisbon, Portugal

https://ss25.ihpcss.org

- Canadian students applied: **137**
- Canadian students selected: **10**
- Returning mentor: **1**
- Instructors sent: Yohai Meiron, James Willis, and Ramses van Zon (funded by SciNet)
- IHPCSS Working group:
  Sarah Huber, Moïra Dion, Ramses van Zon, and Catherine Di Vita
- Reviewers: Sarah Huber, Moïra Dion, Yohai Meiron, Alex Razoumov, James Willis, Sarah Cameron-Pesant, Bruno Mundim, Vladimir Slavnic, Marco Saldarriaga, Chris Want, Jerry Li, Matthew Smith, Paul Preney, Chris Loken, Jane Fry

IHPCSS 2026: Perth, Australia

- Reviewers: Yohai Meiron, Alex Razoumov, James Willis, Bruno Mundim, Vladimir Slavnic, Marco Saldarriaga, Chris Want, Jerry Li, Matthew Smith, Gurpreet Matharoo Chris Loken, Olivier Fisette, Ross Dickson, Patrick Mann

https://ss26.ihpcss.org

- Canadian students applied: **232**
- Canadian students selected: **10**
- Returning mentor: **1**
- Instructors to be sent: Sarah Huber, Alex Razoumov, Ramses van Zon
- Alliance representative: Eduardo Fuenmayor
- IHPCSS Working group: Sarah Huber, Julie Faure-Lacroix, Ramses van Zon, and Tania Tan

Evolving content

- The main focus of the IHPCSS remains large-scale HPC.
- Technical content is reviewed annually.
- The IHPCSS has changed significantly since 2010:
  - GPU programming is now a standard component.
  - Big data/machine learning/AI component continues to grow.
  - There's even an Python for HPC session.
  - But, intentionally, it is not becoming an AI conference.
  - The mentoring component is unique and well-established.
- Every year, substantial effort goes into collecting feedback, assessing the skill landscape and updating the technical content.

Scott Callaghan et al. 2025. Fifteen Years of International HPC Summer School. In PEARC '25, ACM, New York, NY, USA, Article 17, 1–8. **https://doi.org/10.1145/3708035.3736011**

IHPCSS Impact

- ~92% found that **participants from other countries** enhanced their learning.
- ~97% rated the **experience successful**.
- The **structured mentoring program** was identified as one of the most unique and valuable aspects.
- The 10 students from Canada get to **network** with ~70 peers and ~30 international experts, including returning mentors.
- The 4 Canada **staff members** get to **network** with ~30 international peers.
- The **Alliance/Canada** gains **exposure** to ~30 international HPC peers.
- **100** students have been sent from Canada over the last 12 years, many moving into leading roles in **academia**, **government**, and **industry**.

Comments from students

"I enlarged my network of peers drastically. Got a good taste of many many tools that will directly help my research."

"Taking answer 'useful knowledge and skills for future work' as a self-evident, I would say that it is motivation for self-development. In this Summer School I met many people that are examples to follow and with whom I would like to work in future. It incredibly ignited me."

"The thing that I definitely took away was the connections I made with not only other students from other countries, but with the instructors as well. Usually after attending these types of workshops and schools, I don't particularly have a desire to keep in contact with people, but at IHPCSS I met numerous people I would love to keep in contact with."

"I think that the summer school provides an amazing networking opportunity for these budding young researchers."

Lessons Learned

- Budgets need healthy margins for rising costs and uncertain participant origins.
- VISAs remain unpredictable; in 2025, two students could not attend because of VISA appointment delays.
- The working group only needs frequent meetings during a few key periods.
- We need more reviewers (or less advertising…).
- Coordinating finances across two organizations is slow and painful; best avoided.

- But it is worth it; demand is very high.
- Hoping for a multi-year training strategy that includes the IHPCSS! (so we may bring it to Canada?)